

I Viswanath

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EDUCATION

National Institute of Technology, Trichy

Bachelor of Technology in Computer Science and Engineering

Tamil Nadu, India

2020 – 2024

OPEN SOURCE CONTRIBUTIONS

Linux Foundation – Linux Kernel Bug Fixing Mentorship (Fall)

Remote

Mentee

September 2025 – November 2025

- Upstreamed 6 Linux kernel patches across `net`, `i2c`, and `mm`, including 3 bug fixes in `net`—2 backported to stable—such as a fix for a race in an Ethernet-over-USB driver.
- Fixed an async race where `set_rx_mode()` could restart the TX queue while `start_xmit()` was active, resulting in double URB submission ([GitHub](#)).
- Currently implementing a maintainer-proposed redesign of the `dev_set_rx_mode()`, moving RX mode I/O out of a spin lock context to eliminate unnecessary busy waiting and improve I/O safety. ([GitHub](#)).
- Maintaining a public repository documenting upstream patches and design discussions ([GitHub](#)).

EXPERIENCE

Centre for Development of Telematics (C-DOT)

Bengaluru, India

Software Engineer

August 2024 – Present

- Adapted platform drivers to enable the embedded software stack to run on the x86 architecture selected for the product, overcoming vendor ACPI configuration limitations.
- Handled I2C and GPIO interactions in the application stack to support board configuration and system diagnostics.
- Developed a custom device driver for a critical component enabling essential router functionality.
- Re-architected the system monitoring stack to improve modularity, readability, and maintainability, simplifying support for future router platforms.

Arcesium

Bengaluru, India

Summer Intern

May 2023 – July 2023

- Developed a new web architecture using Spring Boot and Kotlin, enhancing performance and reliability of the integration subsystem.
- Implemented direct database access for plain queries and integrated a secret vault mechanism to ensure secure handling of customer data.
- Optimized API performance, achieving a **3.5× reduction** in average response time.

PROJECTS

Game Boy Emulator (GGBE) C++, SDL2

Personal Project

- Implemented a cycle-accurate CPU core and hardware subsystems for the original Nintendo Game Boy in C++.
- Integrated SDL2 for real-time graphics and input handling.
- Validated correctness and timing accuracy using open-source test ROMs and homebrew benchmarks.
- [GitHub Repository](#)

SKILLS

Programming: C, C++, Embedded C, Python, Shell Scripting

Systems: Linux Kernel, Device Drivers, Kernel Modules, Interrupt Handling

Hardware Interfaces: I2C, UART, GPIO

Debug & Tools: GDB, strace, ftrace, printk, QEMU, buildroot

Build & Integration: Embedded Systems, make, git, Cross Compilation, CI/CD